



let's connect

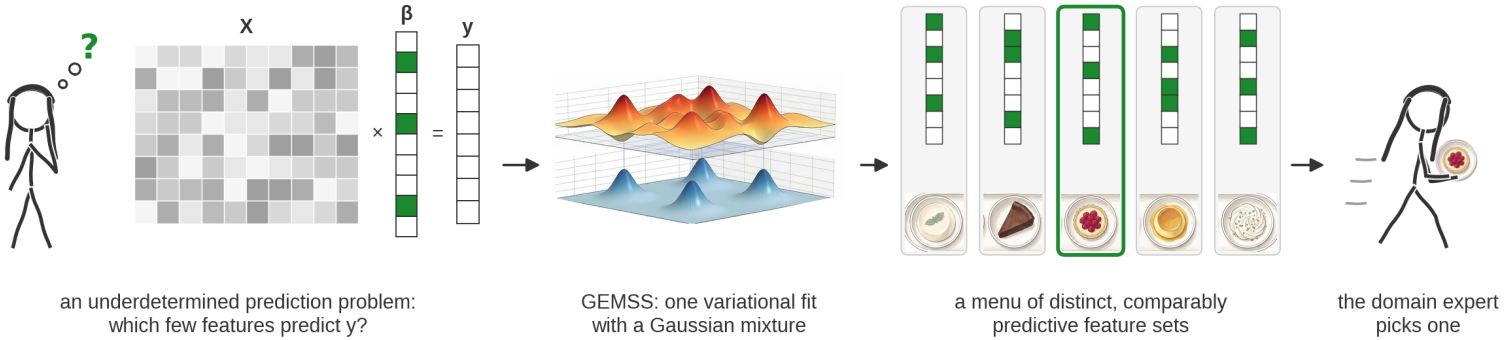
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Underdetermined problems have multiple sparse solutions. GEMSS serves you the menu.



Few samples, too many features

Tens to hundreds of samples against hundreds to thousands of candidate features: omics, lab measurements, sensor data...



Expensive measurements

Every sample costs lab time, money or a patient visit, so the data will stay limited. Get all the insight out of what you have using Bayesian statistics.



The obvious solution is unusable

The dominant signal in your data is not what you are looking for but it cannot be filtered away. See the alternatives.



Domain expertise and human judgement are essential

Statistics is not everything. Pick the set that is best supported by domain knowledge, easiest to apply or cheapest to measure. You prioritize the options.

GEMSS

= Gaussian Ensemble for Multiple Sparse Solutions

The starting point: underdetermined problem, generalized linear model.

The algorithm:

1 Prior: Structured Spike & Slab (Andersen et al., NeurIPS 2014), exactly D nonzeros, mixing over size- D supports A :

$$\pi(\beta) = \frac{1}{|A|} \sum_{A \in \mathcal{A}} \prod_{j \in A} \pi_{\text{slab}}(\beta_j) \prod_{j \notin A} \pi_{\text{spike}}(\beta_j)$$

2 Approximation: a mixture of K Gaussians, one component per mode:

$$q(\beta) = \sum_{k=1}^K \alpha_k \mathcal{N}(\beta; \mu_k, \text{diag } \sigma_k^2)$$

3 One objective, one fit: maximize ELBO over mixture parameters by SGD:

$$\mathcal{L} = \mathbb{E}_q[\log p(y | X, \beta)] - \text{KL}(q \| \pi)$$

4 Get solutions from components: one component $\Rightarrow$ one candidate solution

- Benchmark – fixed solution size:** get D features with largest $|\mu_k|$
- Practice – variable solution size with outlier detection:** get features whose $|\mu_k|$ are z-score outliers (e.g. 2.5 STD).

Diversity mechanisms:

- Built-in:** the mixture entropy in the ELBO repels components from a shared mode.
- Optional:** penalty on Jaccard similarity allows for direct user control.

A novel benchmark

Generate problems with multiple known solutions, evaluate recovery

Why: prediction metrics alone cannot tell if practically usable solutions were found.

Generator:

- Rank- r latent mechanism plants several solutions that are equally predictive by construction ($R^2 \sim 1$).
- Parameter *overlap* defines solutions' shared core of features.
- Signal and noise features have matched variance.
- Optional: Gaussian noise, missing values, class imbalance.
- Binarize y for classification.

Evaluation metrics:

- Scored on the returned supports $S_1, \dots, S_m$, not on prediction.
- Union- F_1 :** are all relevant features recovered?
 $\rightarrow F_1$ between the *pooled* returned and planted features:

$$\text{union-}F_1 = F_1 \left(\bigcup_k S_k, \bigcup_j S_j \right)$$

- Dissimilarity: how distinct are the solutions?**

$\rightarrow$ Mean pairwise 1 – Jaccard similarity

$$\text{dissim} = \frac{2}{m(m-1)} \sum_{a < b} \left(1 - \frac{|S_a \cap S_b|}{|S_a \cup S_b|} \right)$$

$\rightarrow 0 =$ all solutions collapsed to one, $1 =$ pairwise disjoint solutions
 $\rightarrow$ A descriptor, not a score: read it against the planted truth.

Why unions, not matched solution pairs? See the blog for more intuition.



Free online app



Try it out yourself



Blog



More human intuition



Full paper



Read the full version



Open source



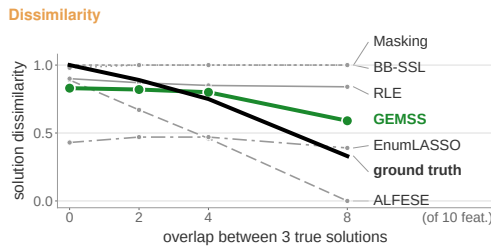
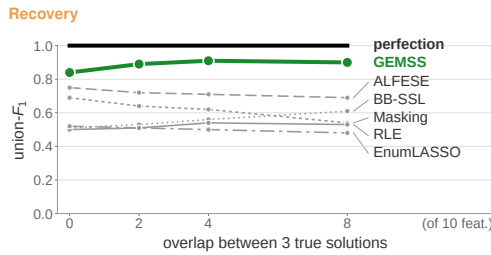
Get code or package
pip install gemss

Experimental results

GEMSS outperforms its competitors on both the benchmark and real data.

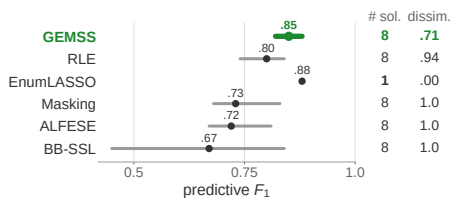
On the benchmark, GEMSS recovers the planted features best at every overlap (union- F_1 .84–.91; next best, ALFESE, $\leq .75$), and the dissimilarity of its solutions follows the true one as the overlap grows. On real data, GEMSS returns 8 distinct solutions (the requirement) with the best predictive F_1 among the methods that do. EnumLASSO collapses to 1 and 6 solutions, respectively. Masking and ALFESE force fully disjoint solutions, BB-SSL nearly so. Only RLE comes close to GEMSS.

Synthetic benchmark: $n = 100$, $p = 200$, 25 runs

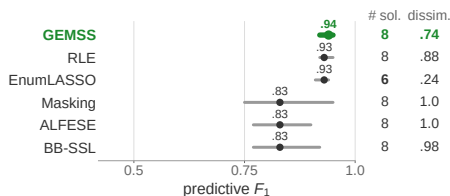


Real metabolomics data

PCOS and preterm birth
($n = 149$, $p = 487$, 8 requested sol.)



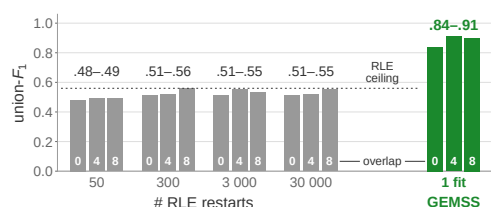
Real data – type-2 diabetes
($n = 132$, $p = 222$, 8 requested sol.)



More restart budget does not help.

On the benchmark, the randomized-LASSO ensemble plateaus within a few hundred restarts, never reaching performance comparable to GEMSS's.

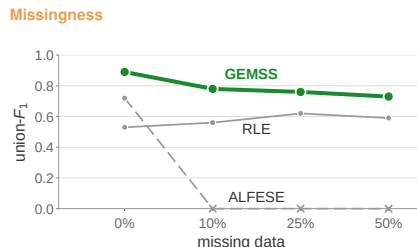
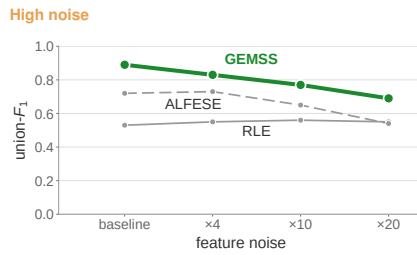
RLE's restart budget vs. GEMSS



GEMSS stays ahead under noise, missing data and $n \ll p$.

We stress-test the 3 best methods on the benchmark problem with added feature noise and missing values. GEMSS keeps the highest recovery metric and degrades gracefully. Missing values are handled natively by computing likelihood only from the present values. ALFESE degrades quickly under noise and cannot run with missing data. In the $n \ll p$ scenarios, GEMSS ranks first in 15 of 18 settings and is fully competitive in the others.

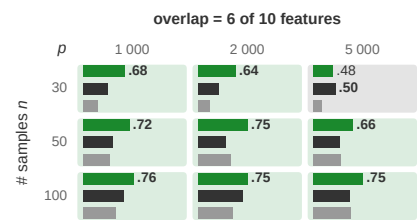
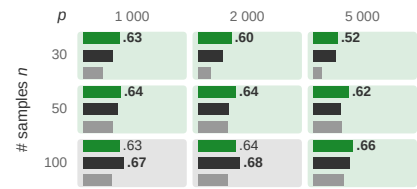
Noise and missing data



High-dimensional $n \ll p$

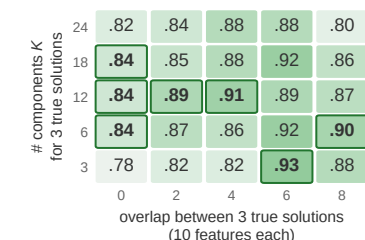
Union- F_1 of GEMSS (green), ALFESE (dark), RLE (light)

overlap = 2 of 10 features



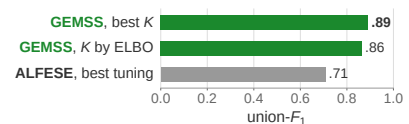
Decouple the number of components from the number of true solutions.

Optimal K vs. overlap (benchmark)



The best number of components K follows the overlap. Disjoint solutions are fitted best with a larger K , which resolves separated posterior modes; for heavy overlap a smaller K is better, avoiding fragmentation of highly correlated supports.

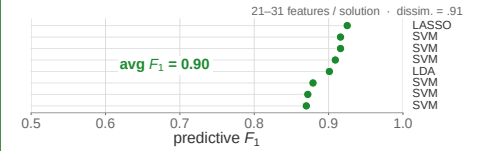
No oracle needed for choosing K . Use the computed objective (ELBO).



Verified: GEMSS is applicable to challenging real-world data.

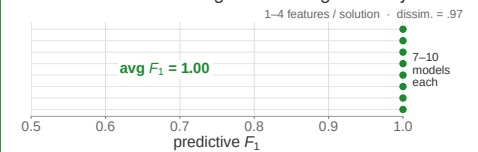
Diabetes biomarkers

- Primary use case $n < p$
- $n = 132$, $p = 222$, ~36% minority class
- ^1H NMR spectroscopy of human urine
- Goal: find metabolites that separate type-2 diabetes from controls
- 8 distinct biomarker sets of 21–31 features, each predictive under nested CV
- Solutions checked against the original study



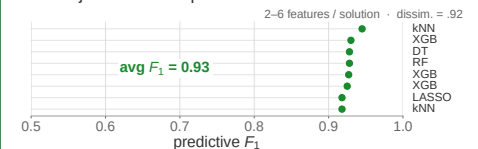
Plant genomics

- Use case with extremely low sample size
- $n = 16$ samples, $p = 41$ identified analytes
- LC–MS metabolomics of *Arabidopsis thaliana*
- Goal: classify wild type vs. double knockout
- Solutions checked against the original study



Food science

- Use case with strongly correlated features ($n = 175 > p = 48$)
- Measured, modelled and exact values; noisy binary labels; minority class ~11%
- Goal: knowledge discovery and a high-recall minority model
- Reproduced findings of prior research and surfaced valid extensions, confirmed by a subject-matter expert



Methods

- **RLE** = Randomized-LASSO Ensemble: stability-selection refits, clustered supports (Meinshausen & Bühlmann, JRSS-B 2010)
- **ALFESE** = Alternative Feature Selection: constrained-optimization wrapper of the Mutual Information selector (Bach & Böhm, IJDSA 2024)
- **BB-SSL** = Bayesian-Bootstrap Spike-and-Slab LASSO (Nie & Ročková, JASA 2023)
- **EnumLASSO** = Enumeration LASSO (Hara & Maehara, AAAI 2017)
- **Masking** = Iterative masking: refit LASSO after removing already selected features

Data sources

- **Diabetes**: MetaboLights, MTBLS1
- **Plant genomics**: MetaboLights, MTBLS2
- **PCOS, preterm birth**: MetaboLights, MTBLS12968
- **Food science**: proprietary commercial data